TEXORA · LECTURE SHEET

Short Staple: Detailed Study of Processing Chemical Staple Fibres and Their Blends with Cotton

Man-made staple arrives clean, uniform and crimped from a cut tow — a completely different starting point from a cotton bale. This sheet covers how that fibre is made to staple length, why its preparation differs from cotton, and how two fibres are blended so that every metre of yarn carries the intended proportion.

YM-4201 · MODERN YARN MANUFACTURING · TOPIC 1 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. What a chemical staple fibre is

A **chemical** (man-made) fibre is produced by forming a polymer into continuous **filament**. Polyester, viscose, acrylic, nylon and modal all begin as filament — an effectively endless strand. A short-staple mill cannot use endless filament, so the filament bundle, called **tow**, is cut or stretch-broken into short lengths that imitate natural staple. The product is **chemical staple fibre**: fibres of controlled length, fineness, crimp and surface finish, supplied compressed in bales.

The contrast with cotton drives everything in this sheet. Cotton arrives variable and contaminated with leaf, seed fragments and dust that must be removed. Chemical staple arrives clean, of almost identical length and fineness, with no trash and no immature fibres. Processing therefore shifts its purpose: not cleaning, but preserving uniformity, controlling static and friction, and — in blends — distributing two fibres so evenly that the yarn behaves as one material.

PROPERTY	COTTON	CHEMICAL STAPLE (E.G. POLYESTER)
Length variation	Wide; short fibres present	Very narrow; cut to one nominal length
Trash and dust	Present; must be removed	Essentially absent
Fineness variation	Considerable, fibre to fibre	Very low; controlled when the filament is spun
Moisture regain	About 7–8% in standard conditions	Polyester about 0.4%; viscose about 11–13%
Static behaviour	Moderate; moisture helps dissipation	Severe in low-regain fibres such as polyester

The regain row explains the central difficulty: a fibre holding almost no moisture cannot leak electrostatic charge away, so polyester charges strongly at every roller and guide (Sections 3 and 4).

2. From tow to staple — cutting, cut length and crimp

2.1 Cutting and stretch-breaking

- **Cutting.** Tow is drawn under tension over blades on a rotating cutter reel pressing against a pressure roll; each revolution cuts one staple length, set by blade spacing and feed. The normal route for cotton-system staple.
- **Stretch-breaking (tow-to-top).** Heated drafting zones break filaments at staggered points, giving a distribution of lengths with tapered ends. Mainly a worsted-system route.

2.2 Choice of cut length

For blending with cotton, staple is normally cut close to the effective length of the cotton it accompanies — commonly about 32–38 mm for medium cottons, and 44–51 mm for longer-staple work.

- **Too short:** little strength contribution, more fly and waste, more hairiness.
- **Too long:** fibres bridge drafting zones, drafting becomes uncontrolled, and fibres lap on rollers; the longest component dictates roller settings for the whole blend.
- **Matched length** lets one set of settings draft both components together and keeps both distributed through the yarn cross-section.

Fineness is matched in the same spirit: a typical blending polyester is about 1.2–1.5 denier per filament (roughly 1.3–1.7 dtex).

2.3 Crimp

A smooth straight filament has almost no fibre-to-fibre cohesion. **Crimp** — a regular zig-zag set into the fibre — supplies the cohesion cotton gets from its natural convolutions: tow is forced into a stuffer box, folds under back-pressure, and the folds are heat-set. Crimp is specified by crimp number and crimp contraction. Too little gives weak, slippery sliver; too much entangles fibres, raises card loading and neps, and makes drafting force excessive. Crimp also contributes bulk and cover to the fabric.

3. Fibre finish

Before baling, the fibre maker applies a **spin finish** — typically well under 1% on fibre weight of lubricating, cohesive and antistatic agents. It sets friction at workable levels, gives the fibre mass

enough cohesion for a stable sliver, and conducts away static. Too little leaves the fibre harsh and strongly charging; too much builds up on cards and rollers and can interfere with dyeing.

4. How processing differs from cotton processing

4.1 Blowroom — opening without cleaning

In cotton the blowroom both opens and cleans. Chemical staple needs opening — dense bales must be fluffed to small, even tufts — but almost no cleaning, because there is almost nothing to remove. Aggressive beating is harmful: it damages crimp, entangles fibres into neps, generates static and fly, and wastes good fibre as droppings. A line for 100% chemical staple therefore uses gentle openers, fewer beating points and reduced or blanked-off grid extraction. A cotton-rich blend line is a compromise between the two.

4.2 Carding

The card still individualises fibres and forms the web, but with little trash to eject its focus is opening without nep formation, preserving crimp and finish, and avoiding damage and roller lapping. Clothing, speeds and settings are adjusted for the synthetic fibre. In blends the card is the last machine at which components can be mixed in tuft form; afterwards, mixing continues only by doubling slivers.

4.3 Static control

Static is the defining nuisance of low-regain fibres: charged fibres repel each other, cling to rollers, produce fly, and cause web breaks and ends down. Controls are used in combination:

- **Humidity** — higher room humidity puts a conductive film on surfaces and lets charge leak away.
- **Antistatic finish** — the conductive component of the spin finish, the defence built into the fibre itself.
- **Machine measures** — grounded metal, static eliminator bars, clean smooth rollers, no insulating covers in fibre-contact positions.
- **Friction discipline** — unnecessary friction points and excessive speeds generate charge faster than defences remove it.

5. Blending — purposes and methods

Fibres are blended to combine properties neither gives alone: polyester contributes tenacity, crease recovery, dimensional stability and easy care; cotton contributes absorbency, comfort and reactive-dye behaviour. A blend is only as good as its **intimacy** (how finely components are mixed) and its **ratio control** (how exactly the target proportion is held). A yarn averaging 65/35 but swinging between 80/20

and 50/50 along its length will dye unevenly — the fibres take different dye classes — and vary in strength.

METHOD	WHERE MIXING OCCURS	INTIMACY	RATIO CONTROL	NOTES
Stock / bale blending (intimate)	Blowroom: weighed tufts fed together	Highest — fibre-to-fibre from the start	Good if weighing is accurate	Carded together; cleaning must suit the cotton
Sandwich blending	Layered batt, fed in vertical slices	High	Good; set by layer weights	Suits small lots and coloured work
Draw-frame blending	Separately carded slivers doubled together at the draw frame	Moderate — sliver-to-sliver, improved by further doublings	Best for exact, adjustable control (Section 6)	Components carded separately under their own best conditions
Roving / yarn-stage combination	Two rovings fed together, or two yarns plied	Lowest — strand-to-strand	Exact in total, poor in distribution	A composite strand, not a true intimate blend

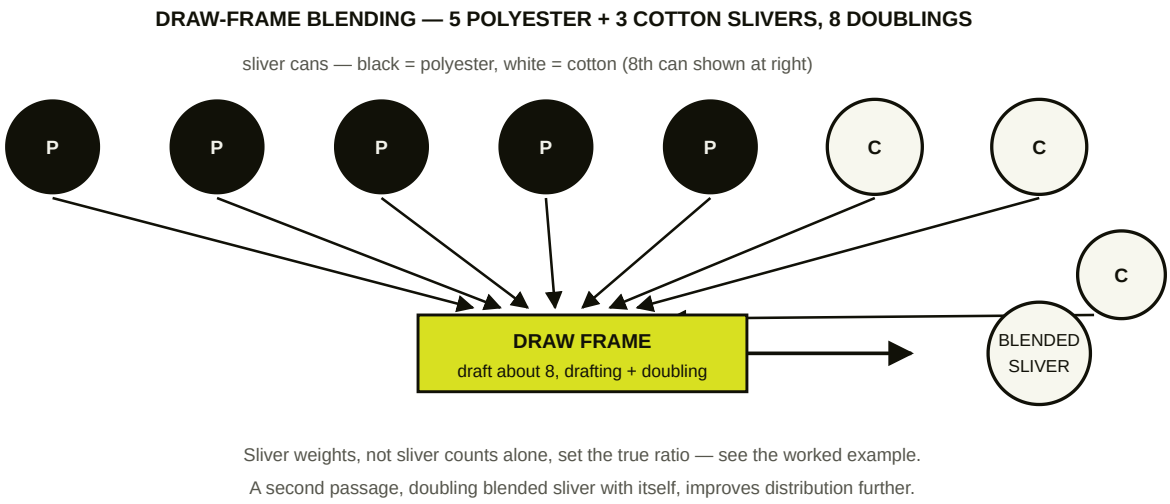


Figure 1 — Draw-frame blending. Eight sliver cans — five polyester and three cotton — feed one draw frame; doubling and drafting combine them into a single blended sliver.

6. Blend-ratio control at the draw frame

With equal-weight slivers the ratio is fixed by sliver numbers alone, in the fractions the doublings allow: with eight doublings, 5 polyester and 3 cotton slivers give 62.5/37.5. Targets such as 65/35 are not eighths, so the standard solution is to keep the doublings and make the components' slivers of *different* weights, because ratio is always a mass ratio:

Component % = (number of its slivers × its sliver weight) / (total mass of all slivers fed) × 100

WORKED EXAMPLE — HITTING EXACTLY 65/35 WITH 8 DOUBLINGS

Feed 5 polyester slivers and 3 cotton slivers, as in Figure 1, but make the polyester sliver slightly heavier: polyester sliver 3.9 g/m, cotton sliver 3.5 g/m.

Polyester mass fed = 5 × 3.9 = 19.5 g. Cotton mass = 3 × 3.5 = 10.5 g. Total = 30.0 g.

Polyester % = 19.5 / 30.0 × 100 = **65.0%**. Cotton % = 10.5 / 30.0 × 100 = **35.0%**. The target is met exactly.

With equal slivers the same 5 + 3 feed would have given 62.5/37.5 — an error of 2.5 percentage points, enough to shift dyeing shade and fabric performance.

Draft check: at a draft of 6 the delivered sliver weighs 30.0 / 6 = **5.0 g/m**, still 65/35.

Control continues after setting: both component slivers are weighed regularly, because card variation makes weights drift. An autoleveller corrects total weight only — it cannot correct ratio, because it cannot tell one component from the other. Ratio is verified by chemical analysis: one component is dissolved away (for polyester/cotton, the cotton with strong sulphuric acid under standard conditions) and the residue weighed. A second draw-frame passage improves distribution; dyeing — polyester takes disperse dyes, cotton takes cellulosic dyes — is the unforgiving audit of any failure.

Glossary

Tow — a large bundle of parallel continuous filaments, the form in which man-made fibre is first produced. **Chemical staple fibre** — tow cut or stretch-broken to short, controlled lengths for staple spinning. **Crimp** — regular waviness set into the fibre for cohesion and bulk. **Spin finish** — a small add-on of lubricant, cohesive and antistatic agents applied by the fibre maker. **Intimate (stock) blending** — mixing components in fibre form before carding. **Sandwich blending** — mixing by layered building and vertical slicing. **Draw-frame blending** — mixing separately carded slivers by doubling at the draw frame. **Blend ratio** — the mass proportion of the components.

Key points

• Chemical staple arrives clean and uniform; preparation opens and preserves it rather than cleaning it.

• Cut length is matched to the cotton and to roller settings — too short weakens, too long defeats drafting.

• Crimp, heat-set in a stuffer box, supplies the cohesion smooth filaments lack; spin finish controls friction and static.

• Draw-frame blending gives the most adjustable, verifiable ratio; intimate blending gives the finest

mixing.

• Ratio is a mass ratio: sliver numbers × sliver weights — different sliver weights make exact targets such as 65/35 reachable.

• An autoleveller corrects total weight, never ratio; ratio is verified by dissolving one component and weighing the residue.

Practice

1. Eight equal-weight slivers are fed at a draw frame, five polyester and three cotton. Calculate the blend ratio and its error against a 65/35 target, in percentage points.
2. A draw frame feeds six slivers: four polyester slivers of 4.2 g/m and two cotton slivers of 3.6 g/m. Calculate the blend ratio.
3. A mill needs 1,200 kg of a 50/50 polyester/cotton blend by intimate blending. How many kilograms of each component must be weighed out?
4. Explain why a blowroom line set correctly for cotton is wrong for 100% polyester staple, naming at least three changes and the harm each prevents.
5. Explain why blending staple is cut close to the cotton's effective length rather than much longer, even though longer fibres give stronger yarn in a 100% yarn.

Answer key

1. Polyester = 5/8 × 100 = **62.5%**; cotton = **37.5%**. Error = 65.0 − 62.5 = **2.5 percentage points** low in polyester.
2. Polyester mass = 4 × 4.2 = 16.8 g; cotton = 2 × 3.6 = 7.2 g; total = 24.0 g. Polyester = 16.8 / 24.0 × 100 = **70.0%**; cotton = **30.0%**.
3. Each component = 1,200 × 50/100 = **600 kg polyester and 600 kg cotton**.
4. No trash exists to remove, so strong beating only damages crimp and creates neps and fly — use gentler openers and fewer beating points; grid extraction drops good fibre — reduce or blank it off; low regain makes static severe — add humidity and antistatic provision.
5. Both components are drafted together under settings that must suit the longest fibres present. A much longer synthetic forces wide settings that cannot control the cotton and causes drafting waves and lapping; matched length lets one setting draft both together.